P 6.1

$$\vec{P} = \vec{D} - \varepsilon_0 \vec{E} = (\varepsilon - \varepsilon_0)\vec{E}$$

$$\vec{M} = \frac{\vec{B}}{\mu_0} - \vec{H} = \left(\frac{1}{\mu_0} - \frac{1}{\mu}\right)\vec{B}$$

Idea: express induced current via $\sigma_{ij} E_j$

$$\vec{J} = -i\omega \vec{P} + i\vec{k}\times\vec{M} = -i\omega(\varepsilon-\varepsilon_0)\vec{E} + i\vec{k}\times\vec{B}\left(\frac{1}{\mu_0}-\frac{1}{\mu}\right)$$

$$\vec{\nabla}\times\vec{E} = -\frac{\partial \vec{B}}{\partial t} \Rightarrow \vec{k}\times\vec{E} = \omega\vec{B}$$

$$\cancel{\vec{\nabla}\times(\vec{\nabla}\times\vec{E}) = -\frac{\partial}{\partial t}\vec{\nabla}\times\vec{B}}$$

$$\cancel{i\vec{k}\times(i\vec{k}\times\vec{E}) = i\omega\, i\vec{k}\times\vec{B}}$$

$$\Rightarrow \vec{k}\times\vec{B} = \frac{1}{\omega}\vec{k}\times(\vec{k}\times\vec{E}) = \frac{1}{\omega}\left(\vec{k}(\vec{k}\cdot\vec{E}) - k^2\vec{E}\right)$$

$$\vec{J} = -i\omega(\varepsilon-\varepsilon_0)\vec{E} + \frac{i}{\omega}\left(\frac{1}{\mu_0}-\frac{1}{\mu}\right)\left(\vec{k}(\vec{k}\cdot\vec{E}) - k^2\vec{E}\right)$$

$$J_i = -i\omega(\varepsilon-\varepsilon_0)E_i + \frac{i}{\omega}\left(\frac{1}{\mu_0}-\frac{1}{\mu}\right)\left(k_i k_j E_j - k^2 E_i\right)$$

$$= \sigma_{ij} E_j$$

$$\sigma_{ij} = -i\omega(\varepsilon-\varepsilon_0)\delta_{ij} - i\frac{k^2}{\omega}\left(\frac{1}{\mu_0}-\frac{1}{\mu}\right)\left(\delta_{ij} - \frac{k_i k_j}{k^2}\right)$$

$$K_{ij} = \delta_{ij} + \frac{i}{\omega\varepsilon_0}\sigma_{ij} \quad \text{(eq. 6.17)}$$

$$= \delta_{ij} + \frac{\varepsilon-\varepsilon_0}{\varepsilon_0}\delta_{ij} + \frac{k^2}{\varepsilon_0\omega^2}\left(\frac{1}{\mu_0}-\frac{1}{\mu}\right)\left(\delta_{ij} - \frac{k_i k_j}{k^2}\right)$$

$$= \frac{\varepsilon(\omega)}{\varepsilon_0}\delta_{ij} + \frac{1}{\varepsilon_0\mu_0}\frac{k^2}{\omega^2}\left(1 - \frac{\mu_0}{\mu(\omega)}\right)\left(\delta_{ij} - \frac{k_i k_j}{k^2}\right)$$

But $c^2 = \frac{1}{\varepsilon_0\mu_0}$ $\Rightarrow$ we obtain (6.18) from the textbook on p. 74

P6.3 (a) $m_e \frac{d\vec{v}}{dt} = -e\vec{E} - \nu_e m_e \vec{v}$

let $\vec{v}(t) = \vec{v}(\omega) e^{-i\omega t}$, $\vec{E}(t) = \vec{E}(\omega) e^{-i\omega t}$

$$\Rightarrow \frac{d\vec{v}}{dt} = -i\omega \vec{v}(\omega) e^{-i\omega t}$$

$$\Rightarrow -i\omega m_e \vec{v}(\omega) e^{-i\omega t} = -e\vec{E}(\omega) e^{-i\omega t} - \nu_e m_e \vec{v}(\omega) e^{-i\omega t}$$

$$(i\omega - \nu_e) m_e \vec{v}(\omega) = e\vec{E}(\omega)$$

$$\vec{J}_{md}(\omega) = -e n_e \vec{v}(\omega) = -\frac{e^2 n_e \vec{E}(\omega)}{(i\omega - \nu_e) m_e} =$$

$$= \frac{i e^2 n_e}{(\omega + i\nu_e) m_e} \vec{E}(\omega) \Rightarrow \sigma_{ij}(\omega) = \delta_{ij} \frac{i e^2 n_e}{m_e(\omega + i\nu_e)}$$

$$K_{ij}(\omega) = \delta_{ij} + \frac{i}{\varepsilon_0 \omega} \sigma_{ij}(\omega) = \delta_{ij}\left(1 + \frac{i}{\varepsilon_0 \omega} \frac{i e^2 n_e}{m_e(\omega + i\nu_e)}\right)$$

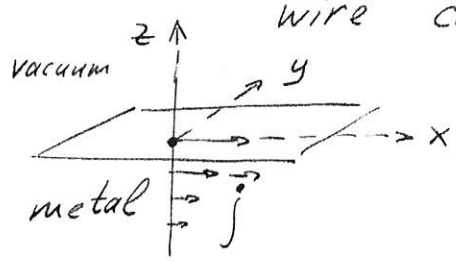
$$= \delta_{ij}\left(1 - \frac{\omega_p^2}{\omega(\omega + i\nu_e)}\right) \quad \text{where } \omega_p^2 = \frac{e^2 n_e}{\varepsilon_0 m_e}$$

(b) $\sigma_{ij}(\omega) = \delta_{ij}\, \sigma(\omega)$ where $\sigma(\omega) = \frac{i e^2 n_e}{m_e(\omega + i\nu_e)}$

$$\sigma(\omega \to 0) = \frac{e^2 n_e}{m_e \nu_e} = \varepsilon_0 \frac{e^2 n_e}{\varepsilon_0 m_e} \frac{1}{\nu_e} = \frac{\varepsilon_0 \omega_p^2}{\nu_e}$$

$$\Rightarrow \eta = \frac{1}{\sigma(\omega = 0)} = \frac{\nu_e}{\varepsilon_0 \omega_p^2}.$$

Problem 3 when $\delta \ll a$ the surface of the wire can be approximated as flat



$\vec{j} = \sigma_0 \vec{E}$

both $\vec{j}$ and $\vec{E}$ are along x.

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = i\omega \vec{B}$$

$$\nabla \times \vec{B} = \mu_0 \vec{j} = \mu_0 \sigma_0 \vec{E}$$ (displacement current is neglected as a small correction at $\omega \ll \omega_{pl}$)

$$\nabla \times (\nabla \times \vec{E}) = i\omega \mu_0 \sigma_0 \vec{E}$$

Since $\nabla \cdot \vec{E} = 0$ (why?) $\quad \nabla \times (\nabla \times \vec{E}) = -\nabla^2 \vec{E}$

$$\nabla^2 \vec{E} = -i\omega \mu_0 \sigma_0 \vec{E}$$

In our setup $\vec{E} = (E(z), 0, 0)$

$$\frac{d^2 E}{dz^2} = -i\omega \mu_0 \sigma_0 E$$ The solution, decaying when $z \to -\infty$ is

$$E(z) = E_0 e^{\sqrt{-i\omega\mu_0\sigma_0}\, z} = E_0 e^{\frac{z}{\delta}} e^{-\frac{iz}{\delta}}, \quad \delta = \sqrt{\frac{2}{\omega \mu_0 \sigma_0}}$$

as $\sqrt{-i} = \frac{-1-i}{\sqrt{2}}, \frac{1-i}{\sqrt{2}} \Rightarrow$ the second value should be chosen

We can now write actual electric field as the real part of the above solution:

$$E(z,t) = E_0 \, \mathrm{Re}\, e^{z/\delta} e^{-iz/\delta} e^{-i\omega t} =$$

$$= \rho_0 j_0 e^{z/\delta} \cos(\omega t + z/\delta), \quad \text{where } \rho_0 = \frac{1}{\sigma_0}$$

we introduced the amplitude of current j_0 at

the surface. It can be related to the total current via

$$I_0 = 2\pi a j_0 \int_{-\infty}^{0} dz\, e^{z/\delta} \cos\left(\frac{z}{\delta}\right) =$$

$$= 2\pi a j_0 \, \mathrm{Re} \int_{-\infty}^{0} dz\, e^{\frac{z}{\delta}(1-i)} =$$

$$= 2\pi a j_0 \, \mathrm{Re} \left(\frac{\delta}{1-i}\right) = \pi \delta a \quad \Rightarrow \boxed{j_0 = \frac{I_0}{\pi a \delta}}$$

Thus,

$$E(z,t) = \rho_0 \frac{I_0}{\pi a \delta} e^{z/\delta} \cos(\omega t + z/\delta)$$

Total Joule heat: per unit length:

$$Q = 2\pi a \int_{-\infty}^{0} dz\, \overline{E(z,t) \cdot j(z,t)} = \frac{2\pi a}{\rho_0} \int_{-\infty}^{0} dz \overline{\left(E(z,t)\right)^2}$$

$$= \frac{2\pi a}{\rho_0} \left(\frac{\rho_0 I_0}{\delta \pi a}\right)^2 \int_{-\infty}^{0} dz\, e^{2z/\delta} \overline{\cos^2(\omega t + z/\delta)}$$

The time average is simply $\overline{\cos^2(\omega t + z/\delta)} = \frac{1}{2}$

and $\int_{-\infty}^{0} dz\, e^{2z/\delta} = \frac{\delta}{2}$, so that

$$\boxed{Q = \frac{\rho_0 I_0^2}{\pi a \delta} = \frac{I_0^2}{\pi a \delta \sigma_0}}$$

Problem 4

$$\vec{j} = \sigma(\omega)\vec{E} = \frac{\sigma_0}{1-i\omega\tau}\vec{E} \qquad \text{(Ohm's law)}$$

$$\nabla\cdot\vec{E} = \frac{\rho}{\varepsilon_0} \qquad \text{(Gauss' law)}$$

$$\frac{\partial\rho}{\partial t} + \nabla\cdot\vec{j} = 0 \Rightarrow i\omega\rho = \nabla\cdot\vec{j} \quad \text{(continuity of charge flow)}$$

Excluding ρ with the help of the last equation $\rho = -\frac{i\nabla\cdot j}{\omega}$ and using it in the second one:

$$\nabla\cdot\vec{E} = -\frac{i}{\varepsilon_0\omega}\nabla\cdot\vec{j} \qquad (*)$$

On the other hand taking $\nabla\cdot$ operation to both sides of the Ohm's law we get

$$\nabla\cdot\vec{j} = -\frac{\sigma_0}{1-i\omega\tau}\nabla\cdot\vec{E} \qquad (**)$$

These two last equations are compatible only if

$$\frac{\sigma_0}{1-i\omega\tau}\times\frac{i}{\omega\varepsilon_0} = -1$$

This is a quadratic equation:

$$\omega\varepsilon_0 - i\omega^2\tau\varepsilon_0 = i\sigma_0 = -i\varepsilon_0\omega_p^2\tau$$

or $\boxed{\omega^2 = \omega_p^2 - \frac{i\omega}{\tau}}$ In the limit of $\omega_p\tau \gg 1$ the last term is small in which ω can be replaced with ω_p to the lowest order:

$$\omega^2 = \omega_p^2 - \frac{i\omega_p}{\tau} \Rightarrow \omega = \sqrt{\omega_p^2 - \frac{i\omega_p}{\tau}} \approx \omega_p - \frac{i}{2\tau}$$

$$\omega' = \omega_p \quad \text{and} \quad \omega'' = -1/2\tau \ll \omega'$$